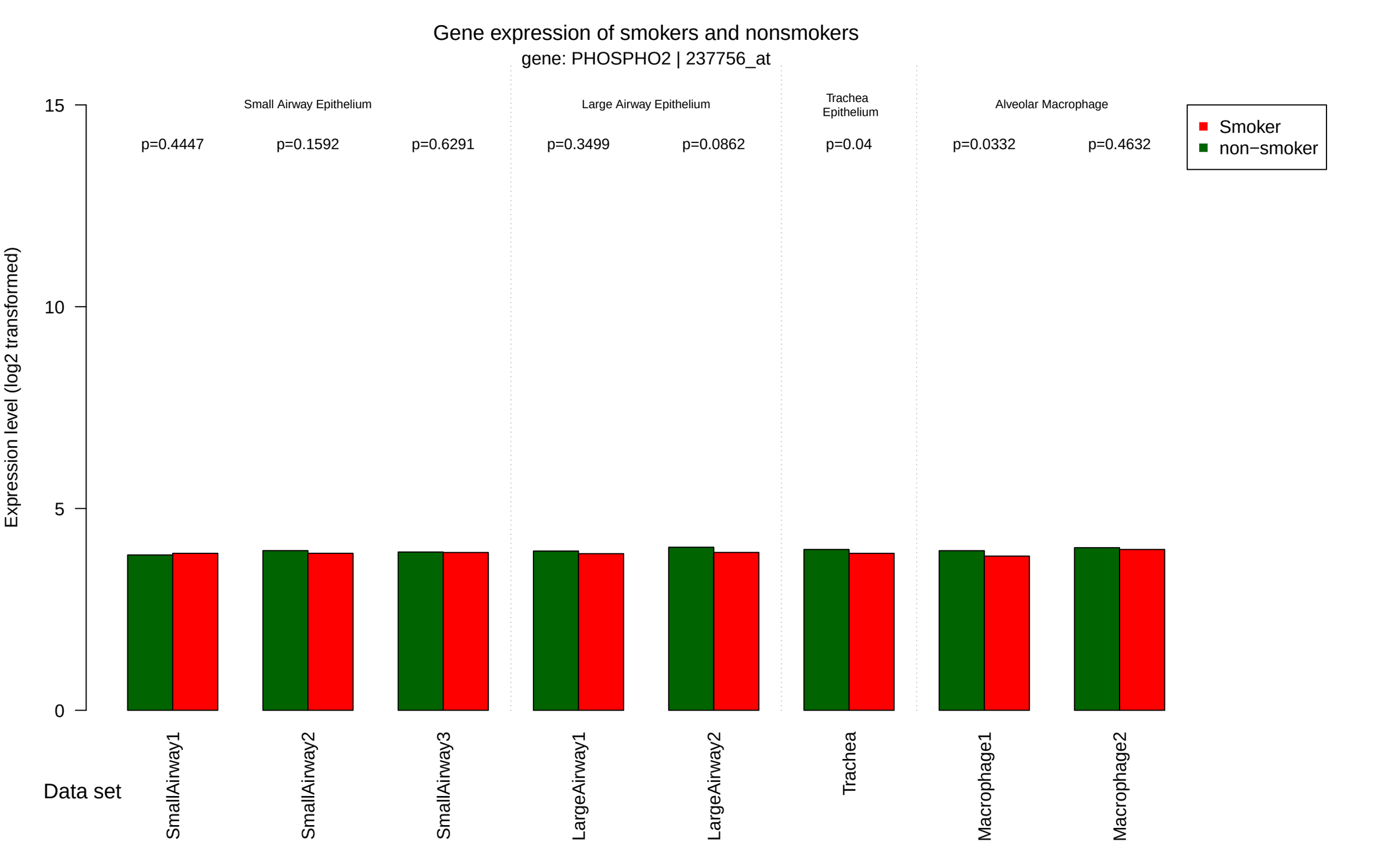
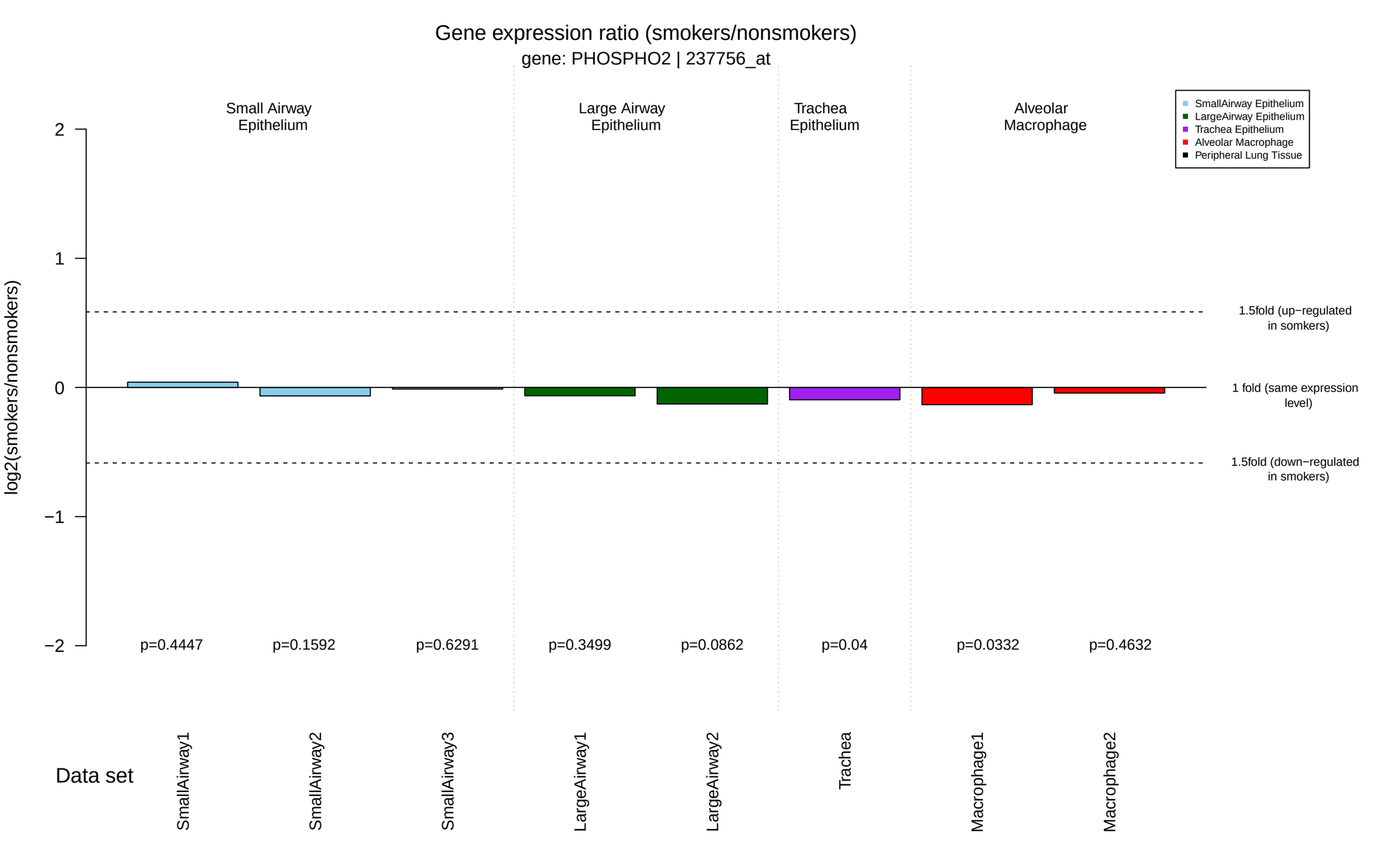
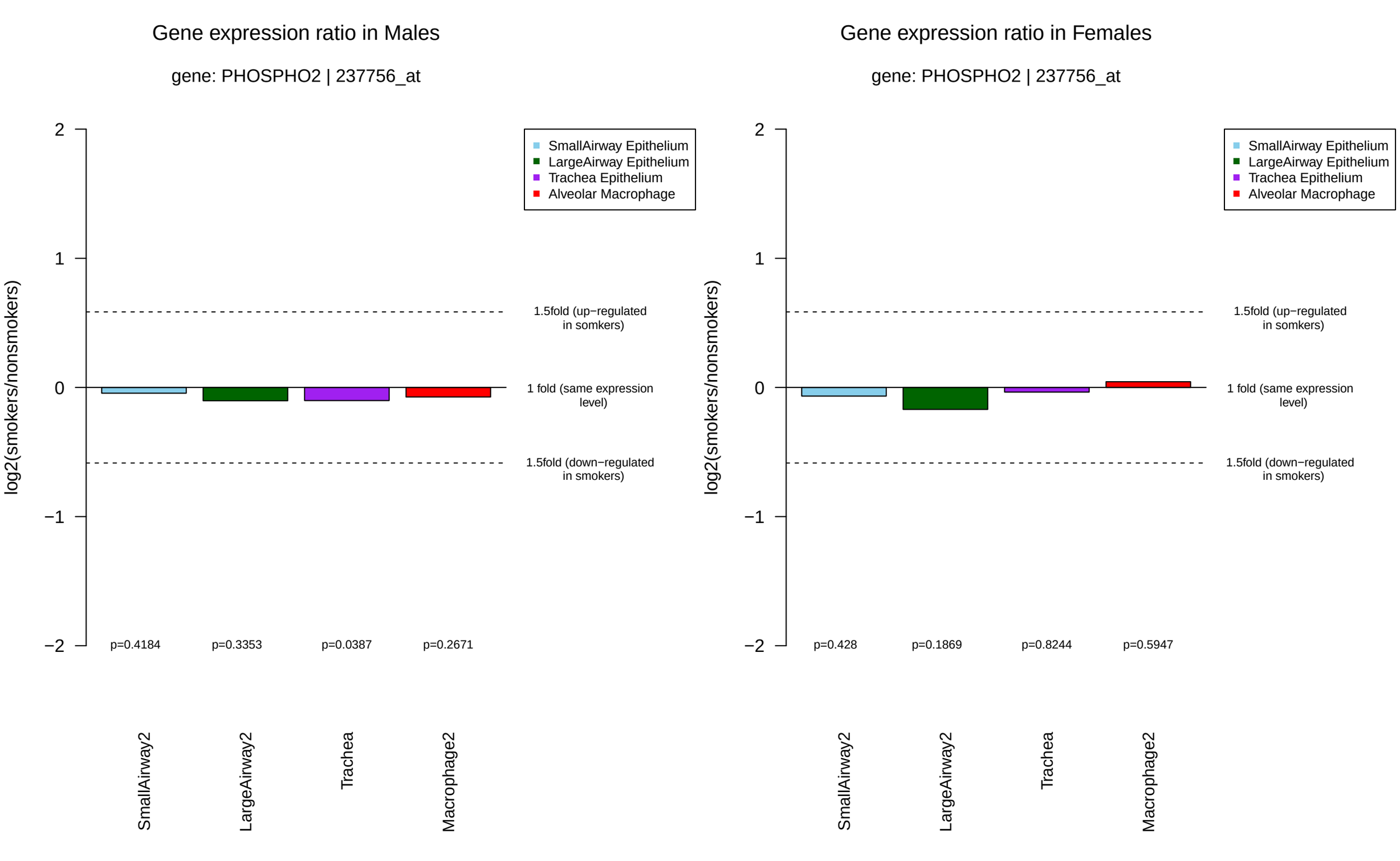
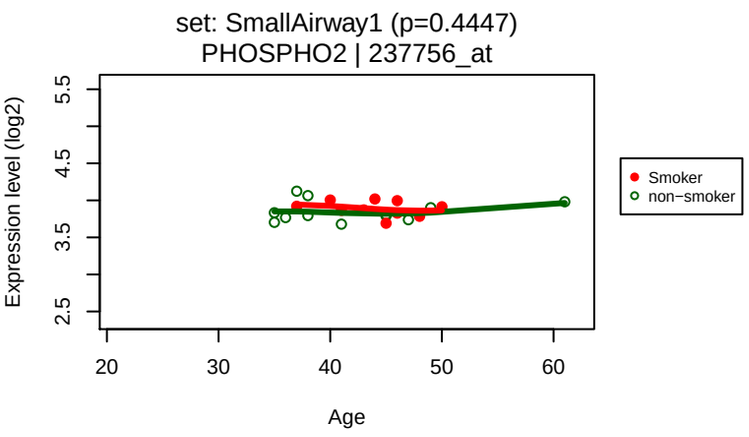
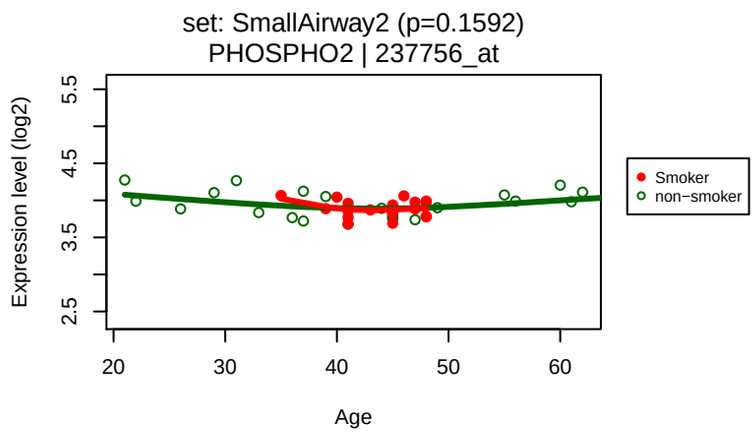
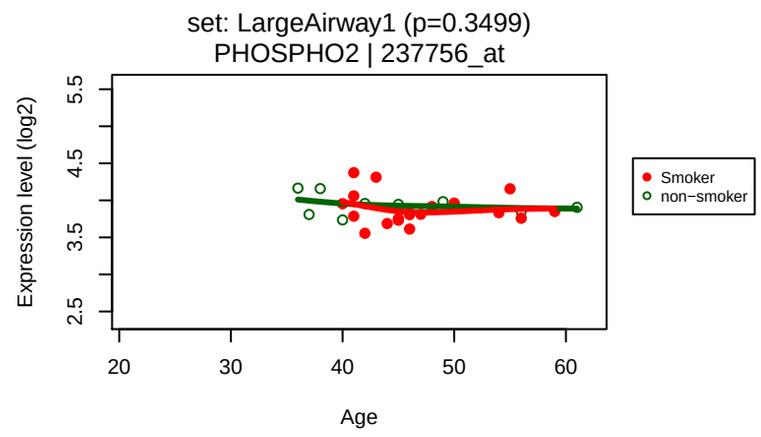

Gene expression of smokers and nonsmokers
gene: PHOSPHO2 | 237756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4447
p=0.1592
p=0.6291
p=0.3499
p=0.0862
p=0.04
p=0.0332
p=0.4632
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

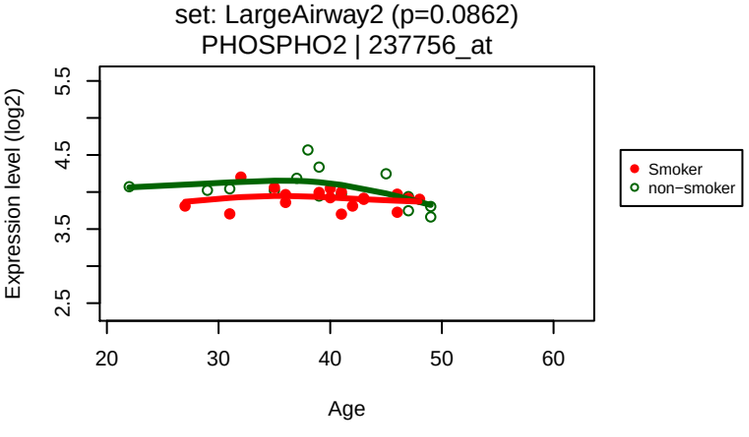

Gene expression ratio (smokers/nonsmokers)
gene: PHOSPHO2 | 237756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4447
p=0.1592
p=0.6291
p=0.3499
p=0.0862
p=0.04
p=0.0332
p=0.4632
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

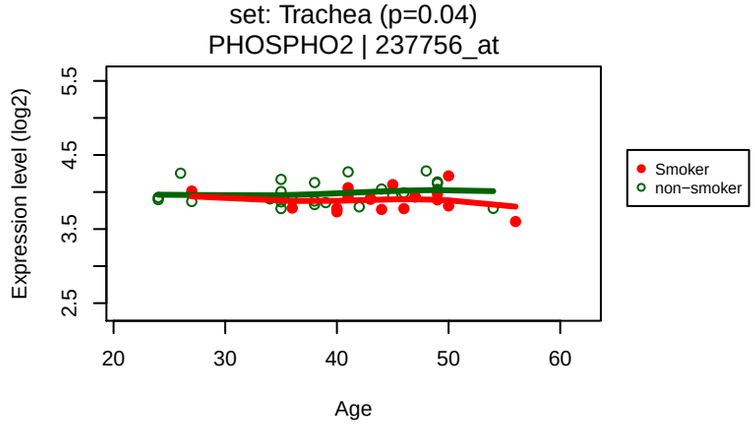

Gene expression ratio in Males
gene: PHOSPHO2 | 237756_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4184
p=0.3353
p=0.0387
p=0.2671
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHOSPHO2 | 237756_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.428
p=0.1869
p=0.8244
p=0.5947
SmallAirway2
LargeAirway2
Trachea
Macrophage2

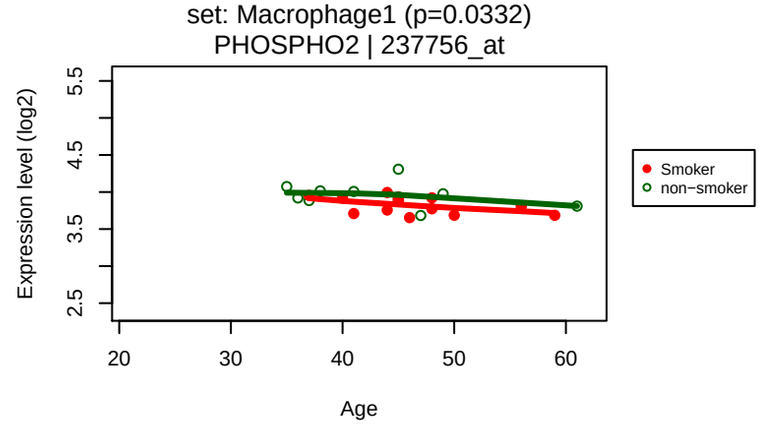

set: SmallAirway1 (p=0.4447)
PHOSPHO2 | 237756_at
Expression level (log2)
Age
Smoker
non−smoker

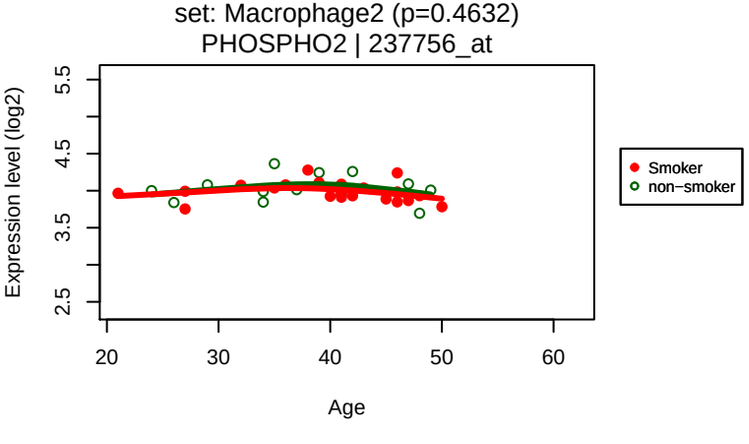

set: SmallAirway2 (p=0.1592)
PHOSPHO2 | 237756_at
Expression level (log2)
Age
Smoker
non−smoker

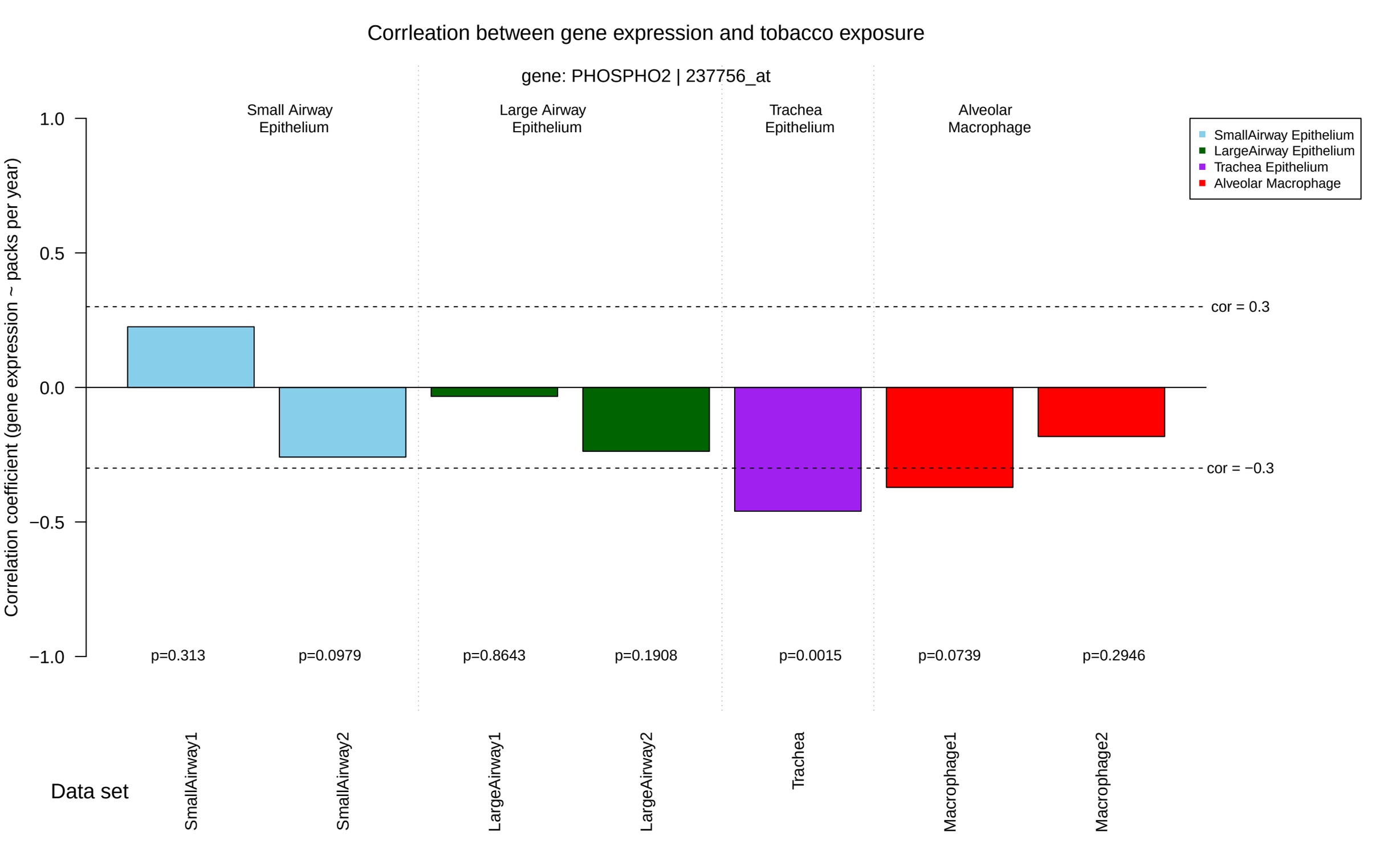

set: LargeAirway1 (p=0.3499)
PHOSPHO2 | 237756_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0862)
PHOSPHO2 | 237756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.04)
PHOSPHO2 | 237756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0332)
PHOSPHO2 | 237756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4632)
PHOSPHO2 | 237756_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHOSPHO2 | 237756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.313
p=0.0979
p=0.8643
p=0.1908
p=0.0015
p=0.0739
p=0.2946
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2